TEXORA · COURSE NOTES · LECTURE SHEET

Development of Surfaces of Prisms, Pyramids, Cylinders and Cones

Unfolding solids onto the flat sheet — the parallel-line and radial-line methods, true lengths, and full worked developments with their governing formulas.

ME-1102 · ENGINEERING DRAWING-I · TOPIC 3 · LEVEL 1 TERM 1

1. What a development is

The **development** of a solid is the shape of its entire surface laid out flat on a plane, every face in its true size and shape, arranged as it would be if the solid were unrolled or unfolded without stretching or tearing. Cut the pattern out of sheet material, fold along the face edges, join the seams — and the solid stands again. Developments are the working drawings of everything made from sheet: ducts, hoppers, cans, lampshades, packaging — and, by direct ancestry, garment patterns. Two rules govern all of it: every line in the development must be a **true length** (developments fail precisely where a foreshortened length is used), and adjacent faces in the development stay adjacent exactly as on the solid.

2. The two great methods

METHOD	IDEA	APPLIES TO
Parallel-line method	The surface is unrolled along lines that stay parallel — the development is built on a stretch-out line equal to the perimeter (or circumference) of the base	Prisms and cylinders — solids whose lateral edges/generators are parallel
Radial-line method	All faces radiate from the apex: the development is drawn with arcs centred on the apex point, using true slant edges as radii	Pyramids and cones — solids whose lateral edges/generators converge to an apex

(A third method, triangulation, covers transition pieces between different shapes, and double-curved surfaces like spheres can only be approximated — neither is exactly developable except by these tricks; this course's four solids are all exactly developable.)

3. Prisms — parallel-line development

A right prism's lateral surface is a row of rectangles, one per base edge, each as wide as that edge and as tall as the prism. Construction: draw the **stretch-out line** of length equal to the base perimeter; mark off the base edges along it in order; erect perpendiculars at each mark of length equal to the prism's height; join the tops — the rectangles stand complete; attach the two polygonal bases (top and bottom) to any convenient faces. Worked dimensions: a square prism of base 30 mm and height 60 mm develops its lateral surface as a strip 4 × 30 = **120 mm wide × 60 mm high**, divided into four 30 mm rectangles, with a 30 × 30 square attached above one rectangle (top base) and below another (bottom base).

4. Cylinders — parallel-line development

A right cylinder unrolls to a single rectangle: its length is the base circumference and its width the cylinder's height:

$$\text{Development of cylinder: rectangle of } (\pi D) \times H$$

Construction: draw the rectangle (in practice the circumference is laid out by dividing the plan circle into 12 equal parts and stepping the chord lengths along the stretch-out line — an approximation of πD, or πD is computed and laid off directly); the circular bases attach as circles tangent to the rectangle's long sides. Worked dimensions: D = 40 mm, H = 60 mm → rectangle (π × 40) × 60 ≈ **125.7 mm × 60 mm**, with the two Ø40 circles attached. A cylinder cut by an oblique plane develops a rectangle whose top edge is a smooth (sinusoidal) curve plotted point by point from the divided plan — the standard exercise on truncated pipes.

5. Pyramids — radial-line development

A right pyramid's lateral surface is a fan of triangles meeting at the apex. Construction: find the **true length of a slant edge** first (in the front view the slant edge is usually foreshortened; rotate it — the rotation method — or compute it: true length = √(vertical height² + distance from apex's plan to the corner²)). With centre at the apex point on paper and radius = true slant length, draw the arc; step the base edges as chords along the arc in order; join each mark to the centre — the triangles stand complete; attach the base polygon to one triangle's free edge. Worked dimensions: square pyramid, base 30 mm, vertical height 50 mm. The apex in plan sits over the square's centre, at half-diagonal distance from each corner: $(30\sqrt{2})/2 \approx 21.2$ mm. True slant edge = $\sqrt{(50^2 + 21.2^2)} = \sqrt{2950} \approx$ **54.3 mm**. The development: four triangles of base 30 mm and equal sides 54.3 mm fanned from the apex, plus the 30 × 30 base square on one free edge.

6. Cones — radial-line development

A right cone unrolls to a **sector of a circle** whose radius is the cone's slant height L and whose arc is the base circumference:

$$L = \sqrt{(r^2 + h^2)}, \quad \text{sector angle } \theta = (r / L) \times 360°$$

Construction: from the apex point draw the sector with radius L and included angle θ (or step the circumference around the arc in 12 parts, as for the cylinder); add the base circle. Worked dimensions — the classic 3-4-5 cone: r = 30 mm, h = 40 mm → L = √(900 + 1600) = 50 mm; θ = (30/50) × 360° = **216**°. The sector of radius 50 mm and angle 216° rolls up exactly into the cone. The same radial logic, with two radii, gives the frustum of a cone — an annular sector between radii L (to the apex) and L_2 (to the cut) — the lampshade and bucket pattern.

7. Workshop and textile notes

- Real patterns add what the pure geometry leaves out: **seam allowances** and laps for joining, and the material's thickness shifts dimensions on tight bends — stated on the drawing as notes.
- Garment pattern cutting is this topic in cloth: the body's surface is developed (with darts absorbing the double curvature a flat sheet cannot take — the triangulation/approximation problem of §2), graded between sizes by stepping measurements exactly as chords are stepped here.
- In the drawing examination, the commonest failure is using an apparent (foreshortened) slant edge instead of the true length — the radial method stands or falls on that one length.

8. Glossary — key terms of this topic

TERM	MEANING
Development	The flat layout of a solid's whole surface in true sizes, ready to cut and fold.
Stretch-out line	The line, equal to base perimeter/circumference, on which a parallel-line development is built.
True length	A line's actual length, as opposed to its foreshortened projection in a view.
Slant edge / slant height	The edge from apex to a base corner (pyramid); the apex-to-base-rim generator length L (cone).
Frustum	The part of a cone or pyramid between the base and a plane cutting off the apex.
Generators	The straight lines that sweep out a cylinder's or cone's surface.

9. Key points and summary

- Development = the surface in true size throughout; parallel-line method for prisms and cylinders (stretch-out = perimeter or πD), radial-line method for pyramids and cones (apex centre, true slant lengths as radii).
- Cylinder → rectangle πD × H; cone → sector of radius L = √(r² + h²) and angle (r/L)·360°; pyramid → fan of true triangles on the true slant edge.
- True lengths are the whole game; frustums use two radii; production patterns add seam allowances as notes.

10. Practice

Written and drawing exercises; brief model answers follow.

1. A triangular prism has base edges 25, 30 and 35 mm and length (height) 70 mm. Give the overall width and height of its lateral development and the shape of each panel.
2. A cylinder of diameter 50 mm and height 80 mm is developed. Give the rectangle’s dimensions.
3. For a cone of base radius 25 mm and vertical height 60 mm, compute the slant height and the sector angle of its development.
4. A square pyramid has base 40 mm and vertical height 45 mm. Compute the true length of its slant edge.
5. Drawing exercise: develop the square prism of §3 (base 30, height 60) full size, attaching both bases, and label the fold lines.
6. Drawing exercise: develop the cone of §6 (r = 30, h = 40) by the 12-part stepping method and check your arc against the computed sector angle.
7. Why must the slant edge’s true length be found before a pyramid can be developed — what goes wrong if the front-view length is used?
8. A lampshade is a cone frustum: top radius 60, bottom radius 100, slant length between rims 120 mm. Describe its development’s shape (radii from the virtual apex must be computed — state the method, using similar triangles along the slant).

Answer key

1. Width = perimeter = 25 + 30 + 35 = 90 mm; height 70 mm; three rectangles 25×70, 30×70, 35×70 side by side (plus the two triangular bases attached to panels).
2. Rectangle = (π × 50) × 80 ≈ 157.1 mm × 80 mm.
3. L = √(25² + 60²) = √4225 = 65 mm; θ = (25/65) × 360° ≈ 138.5°.

4. Half-diagonal = $(40\sqrt{2})/2 \approx 28.3$ mm; slant edge = $\sqrt{(45^2 + 28.3^2)} = \sqrt{(2025 + 800)} = \sqrt{2825} \approx 53.2$ mm.
5. Self-check: strip 120 × 60 in four 30 mm panels; one 30×30 square above a panel, one below another; fold lines are the panel divisions (drawn thin/chain).
6. Self-check: radius 50 mm throughout; 12 chords stepped from the Ø60 plan circle; measured sector angle should come to ≈ 216°.
7. In the front view the slant edges (except any parallel to the view plane) are foreshortened — shorter than reality. Using them makes every triangle too small: the cut pattern would not meet around the base and the folded solid would be undersized and distorted.
8. An annular sector: extend the frustum to its virtual apex by similar triangles along the slant (the full cone's slant L satisfies $r_{bottom}/L = r_{top}/(L - 120)$); the development is the sector of radius L and angle $(100/L)\cdot 360°$ with the inner sector of radius $(L - 120)$ removed — the standard two-radii frustum construction.
